Liya Zhang

Female · 45 · 22 years 1 month experience · Undisclosed · Beijing · Bachelor · Masses

13810478996

liyazhang@163.com

13810478996

Self-description

• Software Engineering Manager, architect, and hands-on technical leader with 20+ years of experience delivering regulated medical devices, medical imaging systems, enterprise software, and AI-enabled engineering solutions.
• Lead globally distributed teams across Product, Program, Mechanical, Electrical, Firmware, Hardware, QA, Manufacturing, and server-platform organizations, translating complex product goals into architecture, milestones, risk controls, and production-ready releases.
• Demonstrated ownership of full product lifecycle delivery, from concept and requirements through architecture, implementation, integration, verification, manufacturing readiness, release, and field support.
• Hands-on AI engineering experience in enterprise RAG, hybrid retrieval, reranking, grounded generation, evaluation, and agentic workflows, with a strong focus on measurable quality, reliability, and practical engineering use cases.

Career Intention

On job, seeking for new job

Medical Device Research and Development Management 60-65k×12Months | Beijing | Medical Instruments, Electronics/Semiconductor/IC, Chemicals for Daily Use

Technology Manager 75-80k×12Months | Beijing | Medical Instruments, Electronics/Semiconductor/IC, Chemicals for Daily Use

Technology Director 60-100k×14Months | Beijing | All

Work Experience

Omnicell 2022/11-To Present

Software Engineering Manager

Responsibilities & Achievements: • Own end-to-end software delivery for ColorTouch, a Windows-based automated medication dispensing platform, including roadmap breakdown, architecture, release planning, staffing, dependency management, technical risk, and quality commitments across multiple releases.
• Serve as the cross-functional software owner across Product, Program, Mechanical, Electrical, Hardware, Firmware, QA, Manufacturing, and globally distributed engineering teams.
• Lead and develop engineers through regular 1:1s, goal setting, performance reviews, career development, and continuous coaching, helping team members improve technical ownership, execution discipline, and cross-functional effectiveness.
• Build a high-accountability delivery culture by clarifying ownership, balancing workload and priorities, removing execution blockers, and using sprint/release outcomes, quality signals, and individual performance to drive continuous improvement.

MedChill V2 | Software Lead, Architect, and Cross-Functional Delivery Owner
• Led the complete software workstream for MedChill, Omnicell's next-generation refrigerated medication management

product, integrating electronically locked refrigerated drawers and bins with established medication dispensing workflows used by hospital clinicians.
• Drove cross-functional and cross-time-zone execution across software, hardware, electrical, mechanical, Product, Program, QA, Manufacturing, and platform teams, aligning scope, interfaces, milestones, dependencies, risks, and release readiness.
• Defined the end-to-end software architecture spanning ColorTouch device software, HardwareService, the HAL-Cabinet hardware abstraction layer, Starbus device communication, SQL Server configuration, and OmniCenter fleet/user management integration.
• Designed and reviewed hardware discovery, registration, addressing, persistence, and command-routing mechanisms that enabled ColorTouch to add, remove, relocate, replace, and operate MedChill drawers without exposing the new physical addressing model to legacy workflows.
• Led the expansion of dozens of ColorTouch UI workflows so medication loading, access, removal, return, inventory, and device-management scenarios could correctly recognize and operate refrigerated drawers and bins.
Selected Customer Escalations – Production Reliability & Patient Safety
• Led resolution of high-priority US customer escalations across Service, Customer Operations, Quality, and Engineering, driving structured triage and root-cause analysis despite incomplete field information, time-zone gaps, and multi-team dependencies.
• Resolved recurring Epson syringe-label printing failures across 10+ customer sites by reproducing field configurations, isolating configuration, media, and driver-related causes, and establishing a standardized restore and troubleshooting procedure validated with Service.
• Investigated a safety-critical CSD dispensing defect by correlating field logs, hardware commands, and code paths; used Codex to accelerate analysis, identified an incorrect workflow branch during multi-medication dispensing, reproduced the field scenario, and verified the fix end to end.
Next-Generation Medication Management Platform
• Contributed to software delivery for Omnicell's next-generation enterprise platform spanning Angular/Nx web applications, backend microservices, and cloud-hosted services.
• Worked with a modern micro-frontend and microservices architecture using Angular, Module Federation, REST APIs, C#/Rust services, and shared platform libraries.
• Built and configured the local development environment and traced application flows across frontend components, APIs, backend services, and supporting infrastructure.
• Worked with applications deployed on AWS/EKS using Kubernetes, Helm, and CI/CD pipelines, gaining practical experience with cloud deployment, scalability, availability, and distributed-system operations.

GE 2015/06-To Present

Staff Software Engineer

Responsibilities & Achievements： - Lead a complex system platform design and framework design;
- Lead the scrum team for daily execution;
- Coach team about high level design and refactoring;
- Coach team about bug investigation and root cause analysis;
- Perform team code review to influence the team about code quality;
- Lead the architecture and design of the E-Manual feature following OO principles;
- Lead the Cyber Security system access control, drive internationally across Beijing and Bangalore teams to deliver technical solution, complete the integration of solution;
- Coach team members about the system architecture and clinical knowledge;
- President award for OEC One VAS project;
- Feature Owner of Cine Playback, responsible for architecture design and implementation, including playing/pause/previous frame/next frame, auto playback, scrolling frames, open/close Cine, managing the imaging pipeline for Cine, adjusting algorithms to adapt to Cine images processing, Cine images presentation;
- Related with Cine Playback, design and implement vascular related functions like subtracted cine, set mask, view subtracted, and also design and implement the CINE set cues and slow motion playback function;
- Feature Owner of Reference Image Hold, this is a very important and valuable function in Seastone VAS product, design and implement in C++ to switch the LIVE and Reference image display, familiar with image display and presentation, and because of its great value;
- Patent is applied for Reference Image Hold;
- Feature Owner of Digital Pen, this is a bright new feature for drawing freely on X-ray images to mark the vessels skeleton to help surgeons to quickly finish the surgery during X-ray is on. As the feature owner, I lead the workflow and architecture design in Java.
- Lead designer of E-IFU: E-IFU is to display the electronic instruction for use on C-Arm system. The most challenging part of design is to display the PDF files and how to validate the documents from the security point of view in our system. Certificate based signatures were imported in our system for this feature so that the system have more security. I made the

solution investigation and design and implementation in Java.
- Familiar with RPC communication using CORBA and Thrift;
- Design and implement the communication between workstation and tablet through thrift framework;
- Design and implementation of Android App on tablet;
- UI design and implementation with Java Swing and OpenGL;
- DB OR mapping with Hibernate;
- Manage code repository as a SCM, making releases, setup Continuous Integration system in Jenkins;

IBA 2006/07-2015/03

R&D · Software engineer

Income：18342RMB × 12Months

Position of reports To：R&D manager

Responsibilities & Achievements：- Develop web application “Patient Data Controller” in ruby on rails, including coding and testing in layers of presentation, controller, and model;
- Participate in software development for system “Treatment Room Control System”, including coding and writing unit test in JUnit;
- Develop the system “Irradiation Data Translation” in Java, implementing the algorithm according to the formula given by physicians, and building a web application using Ruby on Rails to provide a method about testing the translation result;
- Develop TRCS (“Treatment Room Control System”) in Java, including coding and writing unit test in JUnit, using technologies like spring, jdbc, ibatis, swing;
- Making software test plan according to project roadmap, designing test cases according to release covered features and requirements;
- Develop automatic testing from zero to covering non-regression test cases in order to setting up non-regression testing for continuous integration daily build in hudson, using QTaste automatic testing framework in java and python, designing test interfaces, maintaining test scripts and data;
- Taking part in developing automatic testing framework features so that it can meets requirements;
- Executing scheduled test cases, filling test result and report, creating bugs in defect management system, validating and testing all the fixed bugs;
- Performing database upgrading test, making sure the old data can be upgraded correctly as required;
- Participating in CCB which is defect management board; make sure the priorities, severity levels, and resolution for each defect found;
- Making releases, and build demo for each release, to make sure old releases can be run at any time;
- Participate in Validation and Verification CCB to get all test reports validated by QA officer;
- Performing management role to help R&D manager through negotiating with requirement engineers, release managers and software developers about release deliveries, development progress, bugs management, and configuration management;
- Lead the test engineers to execute all the test cases, including splitting test tasks, setting up testing environment, directing testing skills;
- Training the developers and test engineers about automatic testing skills, guiding developers to develop software with higher testability.

IBA 2006/07-2008/12

software engineer

Responsibilities & Achievements：- Develop web application “Patient Data Controller” in ruby on rails, including coding and testing in layers of presentation, controller, and model;
- Participate in software development for system “Treatment Room Control System”, including coding and writing unit test in JUnit;
- Develop the system “Irradiation Data Translation” in Java, implementing the algorithm according to the formula given by physicians, and building a web application using Ruby on Rails to provide a method about testing the translation result;
- Develop part of system TRCS (“Treatment Room Control System”) in Java, including coding and writing unit test in JUnit, using technologies like spring, jdbc, ibatis, swing;

AIT 2004/02-2006/06

software engineer

Responsibilities & Achievements: - Researching on protocol stack, service specifications and standards released by ISO, ITU-T, IEC and ICAO, giving the protocol implementation conformance statement(PICS) for aeronautical telecommunication network router and message handling system, and taking part in the international operational testing;
- Performing development in C programming language and taking responsibility for unit test, familiar with gdb debugging, and having good skills in writing bash shell scripts;
- Write interface and integration testing plan and test cases, performing integration testing and filling testing result and test report;

Project Experience

OEC One Yao project 2019/01-2019/12

Senior Software Engineer

Description: This project is to add detector function to the system so that image quality is better.

Responsibilities: - Feature Owner of Digital Pen, this is a bright new feature for drawing freely on X-ray images to mark the vessels skeleton to help surgeons to quickly finish the surgery during X-ray is on. As the feature owner, I lead the workflow and architecture design in Java.
- Lead designer of E-IFU: E-IFU is to display the electronic instruction for use on C-Arm system. The most challenging part of design is to display the PDF files and how to validate the documents from the security point of view in our system. Certificate based signatures were imported in our system for this feature so that the system have more security. I made the solution investigation and design and implementation in Java.

Vascular Project 2017/06-2018/12

Lead Software Engineer

Description: The project is to add vascular functions in current system.

Responsibilities: - Feature Owner of Cine Playback, responsible for architecture design and implementation, including playing/pause/previous frame/next frame, auto playback, scrolling frames, open/close Cine, managing the imaging pipeline for Cine, adjusting algorithms to adapt to Cine images processing, Cine images presentation;
- Related with Cine Playback, design and implement vascular related functions like subtracted cine, set mask, view subtracted, and also design and implement the CINE set cues and slow motion playback function;
- Feature Owner of Reference Image Hold, this is a very important and valuable function in Seastone VAS product, design and implement in C++ to switch the LIVE and Reference image display, familiar with image display and presentation, and because of its great value;

Achievement: - Patent is applied for Reference Image Hold
- President award for OEC One VAS project

TRCS factory testing 2009/01-2015/03

software test engineer

Description: TRCS factory testing is to test Teatment Control System before delivering it to the proton therapy sites so that the software has good quality and accelerate the speed of installation for the whole proton therapy system.

Responsibilities: -Making software test plan according to project roadmap, designing test cases according to release covered features and requirements;
-Develop automatic testing from zero to covering non-regression test cases in order to setting up non-regression testing for

continuous integration daily build, using QTaste automatic testing framework in java and python, designing test interfaces, maintaining test scripts and data;
-Taking part in developing automatic testing framework features so that it can meets requirements;
-Executing scheduled test cases, filling test result and report, creating bugs in defect management system, validating and testing all the fixed bugs;
-Performing database upgrading test, making sure the old data can be upgraded correctly as required;
-Participating in CCB which is defect management board; make sure the priorities, severity levels, and resolution for each defect found;
-Make releases, and build demo for each release, to make sure old releases can be run at any time;
-Participate in Validation and Verification CCB to get all test reports validated by QA officer;
-Performing management role to help R&D manager through negotiating with requirement engineers, release managers and software developers about release deliveries, development progress, bugs management, and configuration management;
-Lead the test engineers to execute all the test cases, including splitting test tasks, setting up testing environment, directing testing skills;
-Training the developers and test engineers about automatic testing skills, guiding developers to develop software with higher testability.

Achievement: - Take over all the testing and delivery activities in the R&D component team;
- Perform automatical testing from 0 to covering all the non-regression test cases;

TRCS development 2006/07-2015/03

software engineer

Description: TRCS is the treatment control system, from which physicist or doctor use it to control the treatment equipment to perform irradiation on patient with cancers.

Responsibilities: - Develop web application “Patient Data Controller” in ruby on rails, including coding and testing in layers of presentation, controller, and model;
- Participate in software development for system “Treatment Room Control System”, including coding and writing unit test in JUnit;
- Develop the system “Irradiation Data Translation” in Java, implementing the algorithm according to the formula given by physicians, and building a web application using Ruby on Rails to provide a method about testing the translation result;
- Develop part of system TRCS (“Treatment Room Control System”) in Java, including coding and writing unit test in JUnit, using technologies like spring, jdbc, ibatis, swing, maven, subversion;

Patient Data Controller web application development 2006/07-2008/12

software engineer

Description: Patient Data Controller is a sub system to let physicist to input patient treatment plan information into database, translating the clinical information into equipment parameters so that the data can be used directly from proton therapy system.

Responsibilities: -Develop web application “Patient Data Controller” in ruby on rails, including coding and testing in layers of presentation, controller, and model;
-Develop the system “Irradiation Data Translation” in Java, implementing the algorithm according to the formula given by physicians, and building a web application using Ruby on Rails to provide a method about testing the translation result;

Achievement: The Irradiation Data Translation and Patient Data Controller systems are now used on most of IBA sites stably.

Education

Beijing Normal University

Bachelor · Computer Science

2000/09-2004/07

Languages

English CET6

French

Mandarin

Additional Info

- 2014.6 ISTQB Advanced Testing Manager Certificate
- 2014.7 ISTQB Advanced Testing Analyst Certificate
- 2018.5~2018.7: Completed "Machine Learning" class on Coursera;
- 2018.11 ~2019.2: Completed "Data Science Essentials-Microsoft: DAT203.1x" class on edx;
- 2019.6~2019.9: Joined PMP training and exam, and PMP certified;
- 2019.12: Joined Certified Scrum Master training and exam, CSM certified;